

termvar, x terminals	$::=$ $\longrightarrow$ $\rightarrow$ $\Rightarrow$ λ $\mapsto$ $\vdash$ $\emptyset$ $\times$ $<:$ $\langle$ $\rangle$	
Γ	$::=$ $\emptyset$ $\Gamma, x : T$	<i>contexts:</i> <i>empty context</i> <i>term variable binding</i>
formula	$::=$ judgement $x : T \in \Gamma$	
t	$::=$ (t) M true false if t_1 then t_2 else t_3 x $\lambda x : T. t$ bind x in t $t_1 t_2$ $[x \mapsto t']t$ M	<i>terms:</i> <i>constant true</i> <i>constant false</i> <i>conditional</i> <i>variable</i> <i>abstraction</i> <i>application</i>
v	$::=$ true false $\lambda x : T. t$ bind x in t	<i>values:</i> <i>true value</i> <i>false value</i> <i>abstraction value</i>
T, S, U	$::=$ Bool $T \rightarrow T'$ (T) M	<i>types:</i> <i>type of booleans</i> <i>type of functions</i>
Jop	$::=$ $t \longrightarrow t'$	<i>Evaluation</i>
Jtype	$::=$ $\Gamma \vdash t : T$	<i>Typing</i>
judgement	$::=$ Jop	

		Jtype
user_syntax	::=	
		termvar
		terminals
		Γ
		formula
		t
		v
		T

$\boxed{t \longrightarrow t'}$ *Evaluation*

$$\begin{array}{c}
\frac{}{\text{if true then } t_2 \text{ else } t_3 \longrightarrow t_2} \text{E_IFTRUE} \\
\frac{}{\text{if false then } t_2 \text{ else } t_3 \longrightarrow t_3} \text{E_IFFALSE} \\
\frac{t_1 \longrightarrow t'_1}{\text{if } t_1 \text{ then } t_2 \text{ else } t_3 \longrightarrow \text{if } t'_1 \text{ then } t_2 \text{ else } t_3} \text{E_IF} \\
\frac{t_1 \longrightarrow t'_1}{t_1 t_2 \longrightarrow t'_1 t_2} \text{E_APP1} \\
\frac{t_2 \longrightarrow t'_2}{v_1 t_2 \longrightarrow v_1 t'_2} \text{E_APP2} \\
\frac{}{(\lambda x : T. t_{12}) v_2 \longrightarrow [x \mapsto v_2] t_{12}} \text{E_APPAbs}
\end{array}$$

$\boxed{\Gamma \vdash t : T}$ *Typing*

$$\begin{array}{c}
\frac{}{\Gamma \vdash \text{true} : \text{Bool}} \text{T_TRUE} \\
\frac{}{\Gamma \vdash \text{false} : \text{Bool}} \text{T_FALSE} \\
\frac{\Gamma \vdash t_1 : \text{Bool} \quad \Gamma \vdash t_2 : T \quad \Gamma \vdash t_3 : T}{\Gamma \vdash \text{if } t_1 \text{ then } t_2 \text{ else } t_3 : T} \text{T_IF} \\
\frac{x : T \in \Gamma}{\Gamma \vdash x : T} \text{T_VAR} \\
\frac{\Gamma, x : T_1 \vdash t_2 : T_2}{\Gamma \vdash \lambda x : T_1. t_2 : T_1 \rightarrow T_2} \text{T_ABS} \\
\frac{\Gamma \vdash t_1 : T_{11} \rightarrow T_{12} \quad \Gamma \vdash t_2 : T_{11}}{\Gamma \vdash t_1 t_2 : T_{12}} \text{T_APP}
\end{array}$$

Definition rules: 12 good 0 bad
Definition rule clauses: 22 good 0 bad